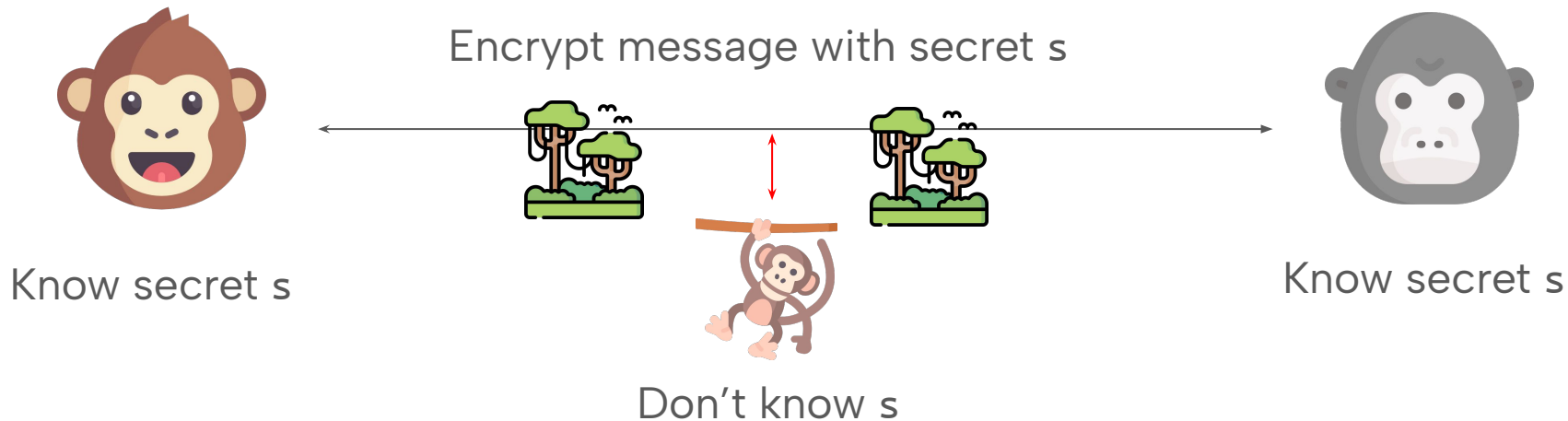
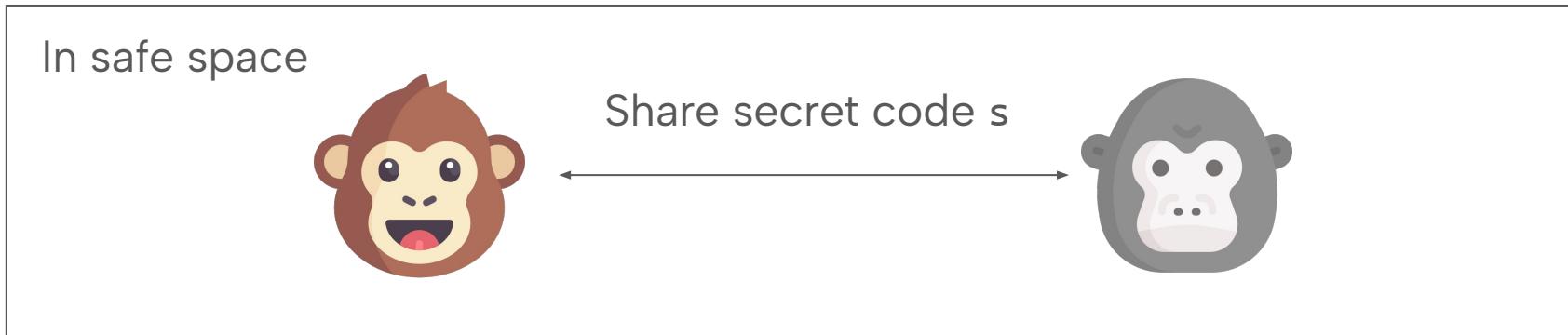


Post Quantum Cryptography

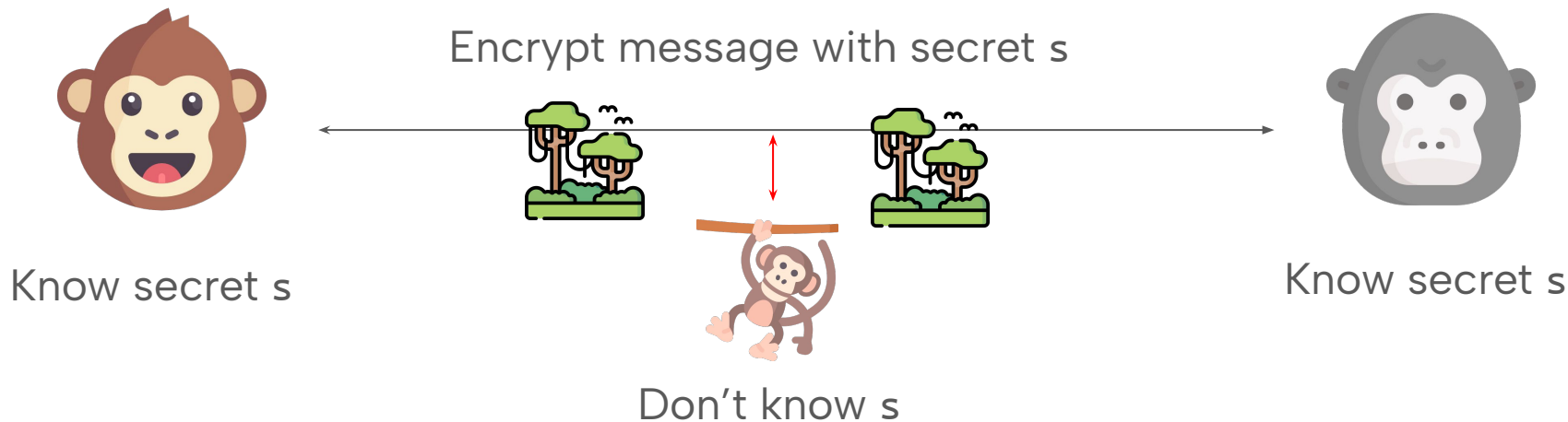
Cyber Skills Level Up UTM

Tan Kel Zin

Symmetric Key Cryptography 1



Symmetric Key Cryptography 2



Pros:

- Efficient scheme exists, e.g AES
- Good security guarantee
- Easier to construct

Cons:

- **Both parties need to share initial secret**

Symmetric Key Cryptography 3

Cons:

- **Both parties need to share initial secret**



Computer in Malaysia

Access Website



Data go through many
intermediate routers



Server in US

How to get share secret key to transmit data securely?

Public Key Cryptography 1



Computer in Malaysia

Send Public Key

Encrypt Message
using Public Key

Generate Public
and Private Key



Server in US

- Only with private key you can decrypt
- Backbone of HTTPS
- **Hard to construct**

Public Key Cryptography 2

Symmetric Key History:

- Greek 500 BC : Scytale (Transposition)
- Roman 50 BC : Caesar Ciphers (Substitution)
- Middle Age 16th Century: Vigenère Cipher (One-time pad)
- World War 2 (1930s): Enigma Machine
- 1997: Data Encryption Standard (DES)
- 2001: Advanced Encryption Standard (AES)

Public Key History

- Early Idea (1969–1973): British intelligence agency GCHQ
- First Construction (1976): Diffie–Hellman Key Exchange
- First Practical System (1977) : RSA (Rivest, Shamir, Adelman)

Public Key Cryptography 3

Importance of Public Key Cryptography

- Backbone of Information Age (HTTPS)
- Secure Messaging

Problem 1 : Inefficient

- Very Inefficient compared to Symmetric Key
- Modern implementation uses Hybrid Encryption
 - Public key to get shared secret
 - Symmetric Key to send data



Problem 2 : Fragile Security

- Diffie-Hellman Key Exchange relies security on **Discrete Logarithm Problem**
- RSA relies security on **Integer Factoring Problem**

Public Key Cryptography 4

Problem 2 : Fragile Security

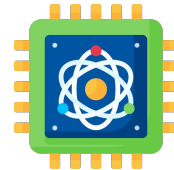
- Diffie-Hellman Key Exchange relies security on **Discrete Logarithm Problem**
- RSA relies security on **Integer Factoring Problem**

- Both problems are assumed to be hard
- They resists attacks for at least 50 years

Shor Invented a **Quantum algorithm** in 1994 that breaks both problems

- Implication: All RSA and DHKE based schemes are completely broken

Fortunately **Quantum Computer** is quite far away from us?



Post Quantum Cryptography 1

Shor's Algorithm inspired **Post-Quantum Cryptography (PQC)**

Goal: Find Public Key Cryptography that is Post-Quantum Safe

Requirements:

- The public key cryptography should run on classical computer
- As efficient as RSA and DHKE
- Resist all known attacks from Quantum Computers

Post Quantum Cryptography 2

Finding PQC turns out to be challenging

After 50 years of work, PQC can be classified by family:

- Lattice-Based
- Code-Based
- Multivariate
- Isogeny-Based
- Hash-Based

This workshop focus

Coding Problem 1

Error Correcting Code



Send Command To Satellite 1000km away



Transmission become noisy



Hello World!



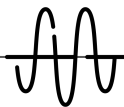
Hnllp Norle!

How to error correct the noisy message?

Coding Problem 2

How to error correct the noisy message?

Hello World!
Hello World!
Hello World!



Zello Woatd!
Hnllp Norle!
Heqio Werld!

Take Majority ↓

Hello World!

Repetition (TCP network)

- Send message Hash, check if message hash
- Incorrect then resend
- Inefficient, only suitable when error is little

Without Repetition?

- Number of characters

Hello World!

Original Message

1H1e2o3l

Additional Information

Coding Problem 3

Every Code need to send some additional data

- More additional data $\rightarrow$ more noise tolerance
- However, more data $\rightarrow$ more inefficient

What makes a code good?

Fix a noise rate, we want both:

1. Efficient Decodable
2. Can tolerate the noise with smallest overhead

QR Code

Packs data and error correcting code into a 2D grid

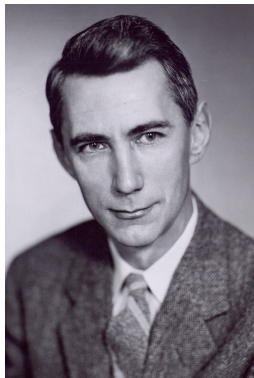
Even when some parts of the code is blurry or get destroyed

It is still possible to recover

Demo [QR Code Generator](#)

Coding Problem 4

Does there exist an ideal code that is both **high rate** and **efficient to decode**?



For every channel, there exists a code that is can achieve theoretically highest rate for the channel

But we don't know how to decode it efficiently

Claude Shannon

Unsolved Problems

We do not know how to efficiently decode some high rate code

Coding Problem 5

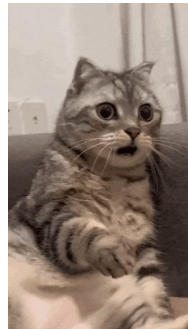
What about **high enough rate code** that are **efficient to decode**?

Famous Efficient Codes with high enough rate

- Hamming Code (1950): Used in Modems
- Reed-Solomon Code (1960): Used in QR codes, Satellite, WiFi 6 & 7
- Low-Density Parity-Check (LDPC) Codes (1960) : Used in 5G network
- Polar Codes (2008): Used in 5G network

More than 50 years of work trying to improve coding
Yet there are still codes that are hard to decode

Turns out to be useful for Cryptography!



Random Code 1

Encode a binary string $s = (0, 1, 1, 0)$

1. Sample a random binary matrix (more rows than column)
2. Encode s by multiplying with A modulo 2

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \longrightarrow m = As^T = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \boxed{\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}}$$

The encoded message we transmit

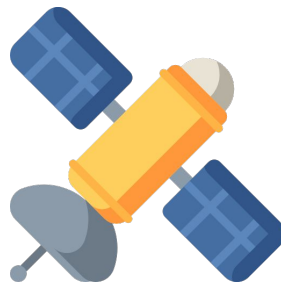
Random Code 2

Both parties know the random matrix

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



Send $A^T s = (0, 1, 0, 1, 0, 0)$



Receive

$(0, 1, \mathbf{1}, 1, 0, 0)$

$$s = (0, 1, 1, 0)$$

$$(0, 1, \mathbf{1}, 1, 0, 0) = \underline{A^T s} + \underline{(0, 0, 1, 0, 0, 0)}$$

How to obtain the original message s ?

Encoded
Message

Introduced Error

Must have small number of 1s

Random Code 3

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Brute Force: Try all possible s

Try $s = (0, 0, 0, 0)$

$$(0, 1, \mathbf{1}, 1, 0, 0) - A^T(0, 0, 0, 0) = (0, 1, 1, 1, 0, 0) \leftarrow$$

Not the correct s because the error is too high

Try $s = (0, 0, 0, 1)$

$$(0, 1, \mathbf{1}, 1, 0, 0) - A^T(0, 0, 0, 1) = (0, 1, 1, 0, 1, 1) \leftarrow$$

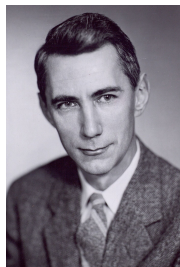
...

Try $s = (0, 1, 1, 0)$

$$(0, 1, \mathbf{1}, 1, 0, 0) - A^T(0, 1, 1, 0) = (0, 0, 1, 0, 0, 0) \leftarrow$$

Correct s !
Small error

Random Code 4



Claude Shannon

Random Code is a code with best rate

Problem: We do not know how to do decoding better than brute force!



Oded Regev

We can do
cryptography with
this hardness!

**Learning With Error
(LWE) in 2005**



Michael Alekhnovich

First construction of
PKE with

**Learning Parity With
Noise (LPN) in
2003**

Learning Parity with Noise (LPN)

1. Sample a random binary matrix A (more rows than column)

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2. Sample a random binary secret s $s = (0, 1, 1, 0)$

3. Sample a random noise e $e = (0, 0, 1, 0, 0, 0)$

LPN Problem:

Given you $(A, As + e)$ Find the s

Michael Alekhnovich 2003 Paper:

If this problem is hard, then we get public key cryptography

Learning With Error (LWE)

1. Choose a prime p , all matrix operation do modulo p $p = 7$
 2. Sample a random matrix A
 3. Sample a random secret s
 4. Sample a random noise e
- $$A = \begin{pmatrix} 3 & 5 & 0 & 2 \\ 1 & 6 & 4 & 3 \\ 5 & 2 & 1 & 6 \\ 0 & 4 & 5 & 1 \\ 2 & 0 & 3 & 5 \\ 6 & 1 & 2 & 4 \end{pmatrix} \quad \begin{aligned} s &= (4, 1, 5, 2) \\ e &= (0, 2, 1, 2) \end{aligned}$$

LWE Problem:

Given you $(A, As + e)$ Find the s

Oded Regev 2005 Paper:

If this problem is hard, then we get public key cryptography

Relevance of LPN/LWE 1

- No known devastating quantum attacks
- Win-win scenario

If LPN/LWE is easy

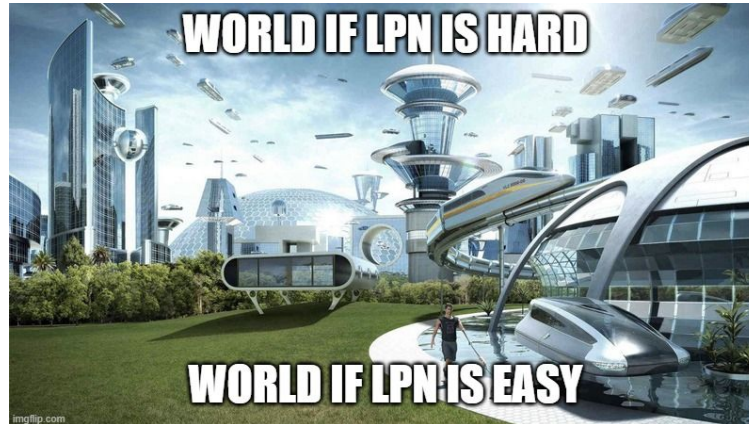
Improvement in error correcting code

- Faster networking

If LPN/LWE is hard

Get security in cryptography

- Public key Cryptography



Relevance of LPN/LWE 2

National Institute of Standards and Technology (NIST)
Standardization of Post-Quantum Cryptography

- FIPS 203 (ML-KEM) : LWE based
- FIPS 204 (ML-DSA) : LWE based
- FIPS 205 (SLH-DSA) : Hash based

LWE also turns out to be very useful for **Homomorphic Encryption**!



Homomorphic Encryption

Is it possible to do computation on encrypted ciphertext?



How to request more food from hoomans



Hey is this a cat? No prompts from cat!

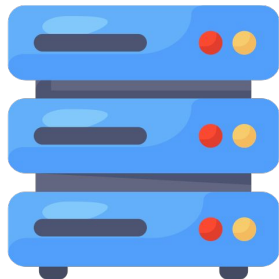
Claude AI



Encrypted prompt



Compute output from encrypted prompt



Decrypt output : Just meow more!

**Server don't know
user prompt**

Decision LPN

Search LPN:

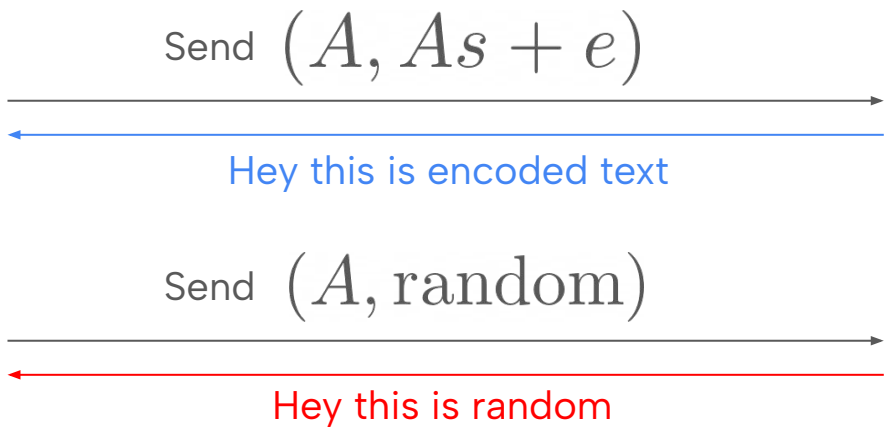
Given $(A, As + e)$ Find the s

Decision LPN:

Distinguish $(A, As + e)$ and (A, random)



Challenger



Distinguisher

Decision LPN

Decision LPN:

Distinguish $(A, As + e)$ and (A, random)

Binary Matrix A:

```
[0 1 0 0 0]
[1 0 1 0 0]
[0 0 1 1 1]
[1 1 0 1 0]
[1 0 1 1 1]
[0 0 1 0 1]
[0 0 0 1 1]
[1 1 0 0 1]
[1 1 0 1 0]
[1 0 1 0 1]
[1 1 1 0 1]
[1 1 1 1 1]
[0 0 0 0 0]
[0 0 0 1 0]
[0 0 0 1 0]
```

```
(1, 1, 1, 0, 0, 0, 1, 1, 1, 0, 1, 1, 0, 0, 0)
```

Is this from $(A, As + e)$, or is this (A, random) ?

Hard to tell!

In fact as hard as the search version!

Decision LPN

Decision LPN Hardness:

No efficient algorithm can distinguish

$$(A, As + e) \text{ and } (A, \text{random})$$

Dual Decision LPN Hardness:

No efficient algorithm can distinguish

$$(A, e^T A) \text{ and } (A, \text{random})$$

Dual Decision LPN

Dual Decision LPN Hardness:

No efficient algorithm can distinguish

$$(A, e^T A) \text{ and } (A, \text{random})$$

```
Error e (noise rate = 0.2000000000000000):  
(0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0)
```

Binary Matrix A:

```
[1 1 0 1 1]  
[1 1 1 1 1]  
[0 1 0 1 1]  
[0 1 0 1 1]  
[0 0 1 1 0]  
[1 0 1 0 1]  
[0 0 1 0 1]  
[0 1 1 1 1]  
[1 1 0 0 0]  
[0 1 1 0 0]  
[1 0 1 0 1]  
[1 1 0 0 1]  
[1 0 1 0 0]  
[1 0 0 0 0]  
[0 1 1 1 1]
```

=

```
Result (e^T * A):  
(1, 1, 0, 0, 1)
```

Alekhnovich PKE 1

Keygen

Public Key: $(A, \underline{As + e})$

Private Key: s

Same



Encrypt

Sample random noise r

Encrypted Message for '0': $(r^T A, r^T (\underline{As + e}))$

Encrypted Message for '1': $(r^T A, \text{random})$

Decrypt

Receive $(r^T A, c)$

Check if $c - (r^T A)s = 0$ Then output '0'
Otherwise output '1'

Alekhnovich PKE 2

Encrypted Message for '0': $(r^T A, r^T (As + e))$

Encrypted Message for '1': $(r^T A, \text{random})$

Decrypt

Receive $(r^T A, c)$

Correctness:

If message is '0', then $c - (r^T A)s = r^T (As + e) - (r^T A)s = r^T e \leftarrow$ Is 0 with high probability!

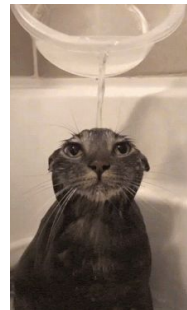
If message is '1', then $c - (r^T A)s = \text{random} - (r^T A)s \leftarrow$ Is 0 with probability half

Noise r and e is sparse

$$\underline{(1, 0, 0, 0)} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

Alekhnovich PKE 3

Your mind right now:



Takeaway 1

What are the things to do in Cryptography now?

Practical Aspect

- Implement the Post Quantum Schemes
- Improve the efficiency of encryption/decryption
- Find attacks on PQC schemes
- Implement advance primitive (Homomorphic Encryption)
- Hardware Attacks on schemes (Side Channel, Fault Injection)

Hardware attack 1

Side Channel Analysis

- Timing, voltage usage, electromagnetic wave of chip leaks information
- Collecting traces can perform key recovery

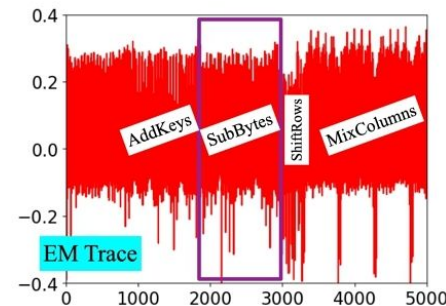
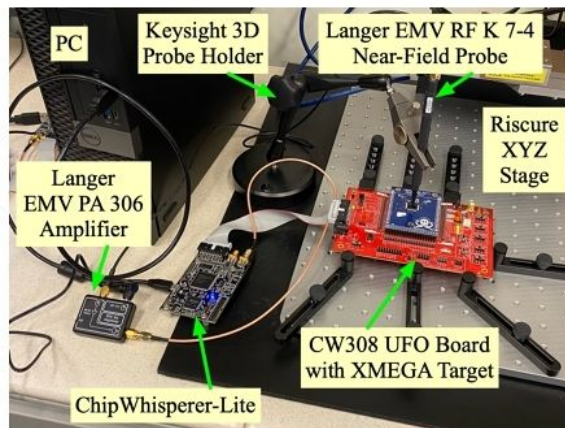


Image Taken from <https://dl.acm.org/doi/fullHtml/10.1145/3678890.3678900>

A Second Look at the Portability of Deep Learning Side-Channel Attacks over EM Traces

Hardware attack 2

Fault Injection

- Inject fault into memory/chips and to change behavior of programs
- A bit flip or skipping an instruction could result in key recovery

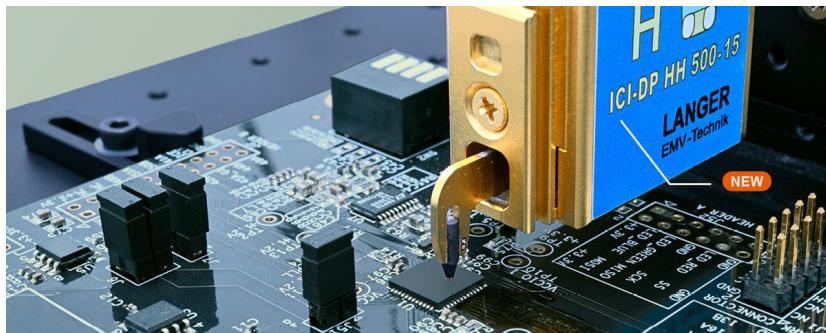


Image from Langer and Keysight

Takeaway 2

What are the things to do in Cryptography now?

Theory Aspect

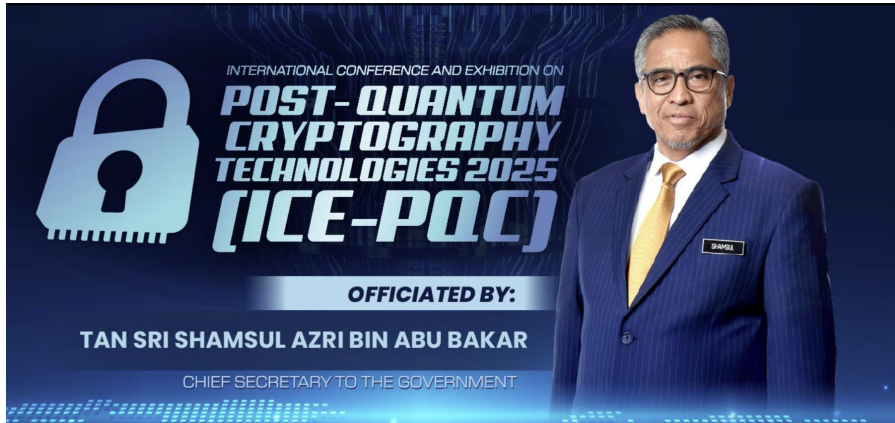
- Find new hardness assumption that can be used for PQC
- Find connections between hard problems
- Theoretical Limitations of Cryptography

**Despite theory proof that LPN/LWE is hard,
Attacks for cryptography scheme still exists**

Cryptography in Malaysia 1

Pusat Teknologi dan Pengurusan Kriptologi Malaysia (PTPKM)

- Actively promote PQC migration
- Raise awareness on digital sovereignty
- PQC sandbox for young talents



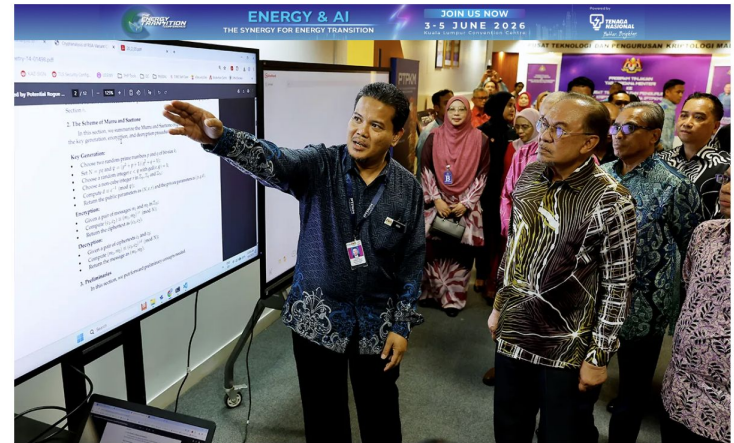
MALAYSIA

PM tinjau Pusat Teknologi dan Pengurusan Kriptologi Malaysia



Bernama

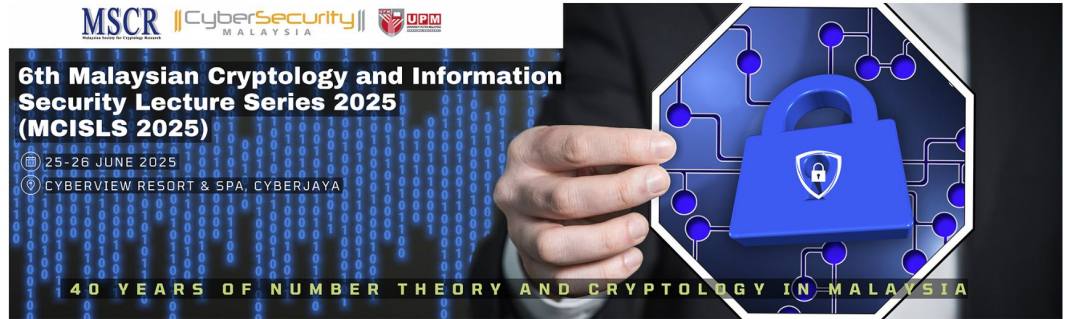
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Cryptography in Malaysia 2

Malaysian Society For Cryptology Research (MSCR)

- Encourage cryptography research
- Promote cryptography education



Malaysian Cryptology and Information Security Lecture Series 2025 (MCISLS 2025)

This Malaysian Cryptology and Information Security Lecture Series 2025 (MCISLS 2025), is another initiative to put forward Malaysian research in the area for public discourse. This 6th series will bring in 8 experts in their related fields for the audience to benefit. [Click here for more](#)



Malaysia Cyber Security Strategy (MCSS) 2020-2024

Malaysia Cyber Security Strategy (MCSS) 2020-2024

MySEAL

Senarai Algoritma Kriptografi Terpercaya Negara

MySeal

National Trusted Cryptographic Algorithm List

MySEAL is a project to develop a portfolio of national trusted cryptographic algorithms. It is a project specifically designed to provide a list of cryptographic algorithms suitable for implementation within Malaysian context that supports Dasar Kriptografi Negara (DKN). While DKN serves as a guiding document for Malaysia to achieve cryptographic sovereignty, MySEAL will support in the scientific areas of cryptography and cryptanalysis.

Thank you